DataLab Start-Up: Ag-GOAT

Prepared by Hilary Choi, Camila Chicatto, Odelyn Xie, Rashmit Shrestha, and Justin Hwang on [May 27, 2026]. Reviewed by [name of any reviewers].

Background: Camila/del

This project was developed through a collaboration between the American Dairy Goat Association (ADGA), the UC Davis Goat Barn, and the UC Davis DataLab. The work focused on the ADGA Linear Appraisal Program, which evaluates dairy goats using a 1-50 scoring system for structural and functional traits related to productivity, durability, and breeding value. These evaluations are used for herd management, breeding decisions, and genetic evaluation research.

Notably, appraisal scores are valuable in breeding goats of the best milk producing capacities. Accurate appraisal scores are imperative for goat breeding because they can help differentiate goats with the highest milking capabilities. This can help identify which goat genotypes - or genetic goat inputs - can produce the best milk production phenotypes - the structural and behavioral goat outputs.

The original research question proposed was whether linear appraisal scores for dairy goat mammary traits could be translated into an anatomically realistic visual model that could improve consistency in appraisal scoring and support appraiser training. The project proposal identified several existing issues within the current appraisal process, particularly for traits that are based purely on appraiser judgement rather than standard measuring utensils such as a ruler. Different appraisers may assign different scores to similar animals based on their unique judgement, and some traits show score distribution compression where only a narrow portion of the scoring scale is commonly used despite visible biological variation.

The original project goals focused on developing a geometric modeling system capable of converting rear udder linear appraisal scores into an adjustable visual representation of the udder. The proposal specifically emphasized rear udder traits, including rear udder height, udder depth, udder arch, medial suspensory ligament definition, teat placement, teat length, and teat diameter. The long-term goal was to investigate whether real-time visual feedback could support calibration, interpretation, and consistency between appraisers. The idea was that real-time feedback would hypothetically help an appraiser adjust their scores and learn what accurate scores look like on the spot. This would help improve the accuracy of the scores while helping appraisers improve their skills.

An early Desmos prototype created prior to the start of the project demonstrated that rear udder traits could be represented using adjustable curves, proportions, and anatomical landmarks. However, the prototype was difficult to scale and lacked a more interactive workflow suitable for broader use.

As the project progressed, the scoped goals became more focused on translating the existing Desmos model into an interactive web based application and developing a functional prototype workflow. The final project goals centered on allowing users to upload an image of a goat, input linear appraisal trait values, and generate an adjustable graphical rear udder model overlaid onto the image for comparison. Additional goals included creating export and import functionality, exploring image overlay workflows, and investigating how visualization tools could support appraisal training and score interpretation outside formal appraisal training sessions.

Field visits to the UCD Goat Barn and discussions with appraisers also influenced the final scope of the project by highlighting how unpredictable real-world appraisal conditions can be. Factors such as the goat's posture, timing relative to milking, bodily movement, and visibility of anatomical landmarks all affected how traits were interpreted and scored.

Partners:

Data Lab Team Members:

- **Project Lead:** Colton Baumler, PhD Candidate, ccbaumler@ucdavis.edu

- **Data Lab Consultant:** Nick Ulle, Senior Statistician, naulle@ucdavis.edu
- **Undergraduate Student Developers:**
    - Camila Chicatto, Team Representative, cchicatto@ucdavis.edu
    - Hilary Choi, Meeting Facilitator, hfchoi@ucdavis.edu
    - Justin Hwang, Scheduling Coordinator, jjhhwang@ucdavis.edu
    - Rashmit Shrestha, Document Manager, rlshrestha@ucdavis.edu
    - Odelyn Xie, Meeting Facilitator, delxie@ucdavis.edu

Goat Barn Team Members:

- **Primary Investigator:** Fauna Smith, Assistant Professor flsmith@ucdavis.edu
- **Goat Facility Manager:** Ben Rupchist, barupchis@ucdavis.edu
- **Student Assistant:** Nora Manring, ezmanring@ucdavis.edu

Approach: Camila

The Data Lab student team worked on developing a web based visualization application from Desmos equations using R and various R packages over the course of ten weeks. Along the way, generative AI models - namely ChatGPT 5 and Claude Opus - were used to facilitate code development.

First, there existed a system of equations on the online graphing calculator Desmos created by Nora Manrig of the Goat Lab. These equations graphed the rear view of a goat, notably graphing the various structures of the mammary (milk producing) system. These equations could be changed through the variables that they relied on, consequently changing the form of the goat udder structures.

Then, the student team worked on translating these equations into code using the coding language R and its respective application R Studio. The team split up the different body part equations so that each member was responsible for translating one set of body part equations into R code, which included the teats, medial suspensory ligament, udder arch, legs, and the udder arch. Then, each member created a function (a reusable, concise piece of code that takes specific variables to perform the same programmed actions each time it is used) that turned the input variables of goat appraisal scores or bodily parameters into a ggplot (a R function/package used to make graphs) visualization of their respective body part. These five different coding files were then combined into one coding file by referencing and calling in the functions from the five original files. This complete visualization then was calibrated to center around a central point, the apex of the pelvic arch, which was treated as the origin for all graphing purposes.

Afterwards, the team worked on turning this code into an application that could be used on appraiser's iPad tablets. This was achieved through integrating the shiny package in R, which is a bundle of functions that help create interactive web applications. The app was created to include the visualization overlaid with an image uploaded by the appraiser, as well as inputs for the appraisal score values that change the visualization to reflect the appraisal scores. Import functionality was added to the app so that basic goat data like their name or ID number could be loaded in and the scores could be associated with a specific goat. Export functionality was also added to the app to export the goat image with the produced overlay as a PNG file, as well as the input appraisal scores that were input at the beginning as a CSV file.

The prototype interface allows a user to first upload an image of a goat and then input linear appraisal trait values into adjustable input fields. Based on these values, the application generates a graphical rear udder model overlaid onto the uploaded image. This allows users to visually compare the generated anatomy against the real goat and evaluate how closely the entered scores match the observed conformation.

Because the generated graph remains adjustable after the initial input, users can continue modifying trait values and immediately observe how the anatomy changes visually. Combined with the application's export functionality, this shifted the tool away from being only a static visualization system and toward functioning as a possible training and calibration aid for appraisers.

Research Outcomes: del + anyone else

The primary outcome of the project was a prototype web application for visualizing dairy goat rear udder linear appraisal traits. Using the existing graphing calculator visualization as a foundation, the team worked toward translating the trait equations and variable relationships into a more interactive and scalable interface. The final prototype is a web based application that appraisers can load on their iPad tablets,then take a picture of a goat, input their appraisal scores, and see the app's produced overlay based off of the inputted scores, and lastly make adjustments as judged before exporting the appraisal score data.

The project also explored additional interface ideas, including image tracing approaches and expanded score comparison systems. Throughout development, the team identified several traits that were relatively straightforward to model geometrically, such as udder depth and teat placement, while others (particularly medial suspensory ligament definition and udder arch shape) remained more difficult to represent consistently without producing unrealistic anatomy.

In addition to the prototype itself, the project produced Shiny workflow components, documentation, and a clearer understanding of the biological and technical challenges involved in translating subjective livestock appraisal traits into computational models.

Discussion: del + anyone else

One of the main challenges of the project was that the mathematical model depended heavily on biological assumptions that were not always fully understood. While some traits translated relatively clearly into geometry, others were much more subjective or difficult to define consistently.

Many of the equations also became more interconnected than expected. Changing one parameter could unintentionally affect multiple other traits or distort the overall shape of the model, making it difficult to fully separate traits into independent components.

Another challenge was scope. The project expanded beyond building a visual model and began including questions about interface design, appraiser usability, image overlays, score comparison systems, and biological accuracy simultaneously. Because of this, it sometimes became difficult to prioritize which parts of the project were most important to complete during the quarter.

The field visit helped clarify many of these issues by showing how variable real-world appraisal can be. Factors like positioning, timing relative to milking, and differences in interpretation between appraisers all affected how traits were perceived. Seeing the appraisal environment in person made the biological reasoning behind several traits much easier to understand and highlighted details that were difficult to recognize from documentation alone.

For future teams working on similar projects, earlier exposure to the animals, appraisal process, and day-to-day workflow of appraisers would likely make development more grounded from the beginning. More frequent communication with the principal investigators and domain experts may also help reduce confusion about trait interpretation and keep technical development aligned with biological reality and project priorities.

Future directions: del + anyone else

One possible future direction discussed during the project was shifting away from relying entirely on manually entered linear appraisal scores and instead allowing users to place anatomical points directly onto an image of the goat. Rather than generating an udder model only from numerical inputs, a point-based system could use traced landmarks and proportions from the photograph itself to generate a more biologically accurate visualization. This could also allow the software to estimate or recommend linear appraisal scores based on the traced anatomy rather than requiring all scores to be entered manually.

This approach may help address some of the score compression and inconsistency issues discussed throughout the project by introducing a more standardized reference system tied directly to anatomical landmarks. It may also provide a more intuitive workflow for newer appraisers who are still learning how traits relate to physical structures. However, a concern with this approach is that it could be used as a crutch by appraisers rather than a training tool. Appraisers may become overly reliant on this feature and avoid the trial and error process necessary to learning how to score more accurate numbers.

Another future direction is expanding the model beyond the rear mammary system. Much of the current work focused specifically on rear udder traits because they are highly important in dairy goat evaluation and already difficult to score consistently. However, similar modeling approaches could eventually be applied to additional areas of linear appraisal and potentially the entire goat.

The team also discussed the possibility of integrating linear appraisal traits with broader scoring systems used during evaluation. Rather than functioning only as an isolated visualization tool, future versions could potentially calculate or assist with final category scores and overall appraisal outcomes within a more complete training or evaluation platform.

The project may also have value as a long-term training and recalibration tool for appraisers. An interactive visualization system could provide a more accessible way for appraisers to compare interpretations, practice scoring, and receive immediate visual feedback outside of formal training events.

Future work would likely require closer collaboration between developers, appraisers, and domain experts to define realistic biological constraints and determine which aspects of appraisal should remain subjective versus standardized computationally.

Literature Cited

- Linear Appraisal Standard Operating Procedure Draft. American Dairy Goat Association. Revised March 2025.
- American Dairy Goat Association Linear Appraisal Program Document.
- Development of a Morphometric Modeling Tool to Improve Human Data Collection Accuracy in Dairy Goat Linear Appraisal

Appendix

- Append the full text of the original start-up proposal
- Include any other figures, links, resources, etc. referenced above

Original Proposal:

Development of a Morphometric Modeling Tool to Improve Human Data Collection Accuracy in Dairy Goat Linear Appraisal

Introduction

Linear appraisal systems are foundational tools in livestock genetic improvement because they convert visual phenotypes into quantitative data used for genetic evaluation. In U.S. dairy goats, American Dairy Goat Association trained appraisers assign 1–50 scale scores to conformation traits. Application of these linear scores should represent the biological range and be approximately Gaussian in distribution, as seen for the measured trait stature (Figure 1A). For this proposal, we focus on mammary system traits scored from the rear of the goat, including rear udder height, rear udder arch, medial suspensory ligament strength, udder depth, teat placement, teat length, and teat diameter. These traits are directly tied to ease of milking, structural durability, mastitis resistance, and lifetime productivity. The reliability of linear scoring therefore directly affects breeding value estimation and long-term herd improvement.

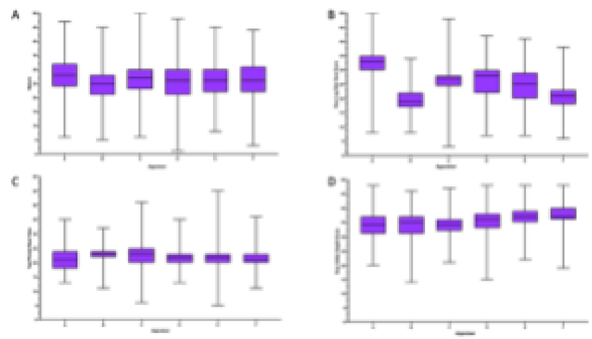

Figure SEQ Figure * ARABIC 1: Examples of data distribution by appraiser*: A. Stature - wide biologic range and gaussian distribution and more range in middle quartiles, B. rear leg side view example of poor inter-appraiser agreement C. Teat placement - compression of biologic range D: Rear udder height – scoring SOP is skewing data to the top of the range.*

Despite standardized operating procedures (SOP) and annual in-person recalibration training, two persistent limitations remain, particularly for non-measured linear traits. First, measurable inter-appraiser variability exists; different appraisers assign meaningfully different scores to similar phenotypes (Figure 1B). Second, score compression is consistently observed, as demonstrated teat placement scores (Figure 1C), despite biologic variation spanning a wide range (Figure 2). In addition, certain traits reveal structural issues in the SOP itself; for example, rear udder height appears skewed toward the upper end of the scale, with few animals scored in the lowest 30–35% of the theoretical range (Figure 1D. Comparisons of photos with assigned scores) further demonstrate that distinct phenotypes may receive similar scores and similar phenotypes receive divergent scores (Figure 2).

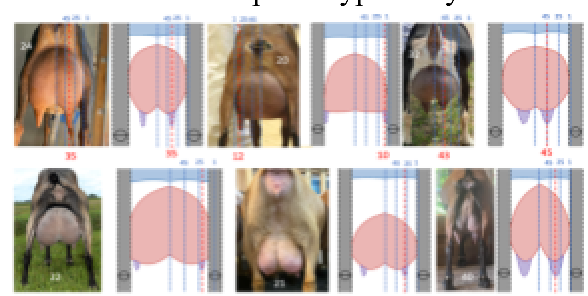

Figure 2: Prototype model using Desmos-graphing calculator scoring teat placement: *Measure scoring system is applied to both the photo of the animal and the generated model of the animal, the number in white in the photo is the number assigned by the appraiser*

Currently, recalibration occurs primarily during annual in-person sessions using live animals, and no quantitative, field-ready calibration instrument exists to support appraisers during routine scoring. We propose to address this gap by developing a parametric visual morphometric modeling system that converts linear trait scores into biologically realistic geometric representations of the rear udder. By providing real-time visual feedback, this system will create a structured perceptual feedback loop that supports self-calibration and more consistent scale utilization, integrating morphometrics, interactive visualization, and data science collaboration to modernize livestock phenotyping.

Collaboration with DataLab is critical to the success of this project because development of the morphometric modeling and real-time visualization system requires expertise in computational modeling and interactive data visualization that does not exist within the Smith Lab. While our team brings deep expertise in dairy goat morphology, linear appraisal systems, and biological interpretation of conformation traits, we do not have the technical capacity to independently build and implement a functional visualization interface. DataLab provides the specialized skills necessary to translate biologically grounded trait definitions into accurate geometric models and a usable interface suitable for controlled pilot testing. This partnership ensures the system is technically sound while remaining biologically meaningful for validation of the proof-of-concept.

Aim 1: Develop a biologically constrained morphometric model that maps 1–50 linear trait scores to continuous geometric representations of rear udder conformation.
Aim 2: Integrate the morphometric model into a prototype visualization tool and conduct a pilot evaluation of its impact on score accuracy, score distribution and inter-appraiser consistency.

Hypothesis:
We hypothesize that a parametric morphometric modeling system providing real-time visual feedback will reduce score compression and inter-appraiser variability in dairy goat linear appraisal. Providing appraisers with an immediate visual representation of their entered scores will promote more consistent interpretation of the SOP and expand effective scale usage.

Approach / Experimental Design:
To address Aim 1 and develop the rear udder model we will need to incorporate all rear udder linear traits and their associated anatomical landmarks as defined by the American Dairy Goat Association Linear Appraisal SOP, including rear udder height (anchored proportionally between the pelvic arch and hock), rear udder arch (center-top curvature at the height reference point), udder depth (vertical distance from the udder floor to the point of the hock, teat placement (lateral position on each udder half measured from the center of teat attachment), teat length and diameter (scaled linear dimensions), and medial suspensory ligament strength (central cleft depth and floor contour). To ensure biologically accurate score-to-geometry mapping, the model will also include structural reference features, specifically the pelvic arch and articulated rear legs with adjustable stance width and hock (calcaneus) to pelvis distance, since these landmarks define proportional scoring boundaries, influence visual perception, and establish the anatomical frame within which rear udder traits are evaluated. An initial exploratory prototype created with graphing-based tools demonstrated feasibility (Figure 2) but lacked reproducibility and scalability. In this model udder contour was be defined by mirrored curves. Rear udder height and arch were map to vertical positioning and curvature parameters. Udder depth was modified by vertical positioning, while teat placement, length, and diameter were be represented by lateral displacement and scaling parameters. Medial suspensory ligament was the most difficult and model manipulation often strayed from biology reality. The protype model can be manually manipulated to mirror the morphology of an udder but lacks the linear score input feature resulting in the udder dimensions.

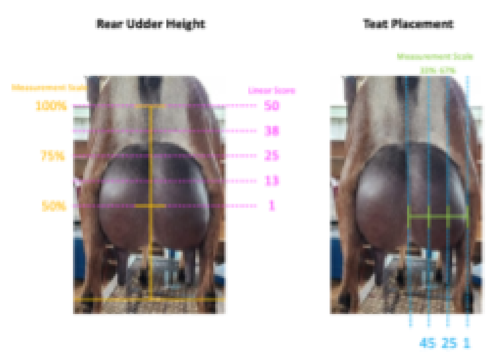

Figure 3: Visual proportion scoring rubric for rear udder height and teat placement. *Yellow and green represent the proportions that correspond with the linear scores (pink and blue) for RUH and TP respectively.*

Each mammary trait will be parameterized according to its standardized method of evaluation. Some traits are a direct measurement and others are proportionate assessment. For direct measurements there are separate scales for standard (Std) and Miniature (Mini) breeds (Table 1). Rear udder height is scored at the start of the lactating tissue relative to body landmarks, using the midpoint between the pelvic arch and hock as a proportional reference (50% = 1; 62.5% =13 ; 75% = 25; 87.5% = 30; 100% = 50) (Figure 3). Rear udder arch is assessed at the same width location as height and scored on curvature at the center-top of the udder, transitioning from pointed (≤19) to rounded (≥20), this trait is the most subjective and scoring methodology is currently being evaluated. Udder depth is evaluated at eye level as the vertical distance between the floor of the udder and the point of the calcaneus (Table 1). Medial suspensory ligament strength is determined by the amount of cleft in the floor of the udder attributable to central support (Table 1). Teat placement is measured from the center of the teat at attachment and scored according to its proportional location on the udder half with one-third corresponding to 45 points and two-thirds corresponding to 25 points (Figure 3). Teat length (base to point) and teat diameter (at attachment to udder floor) are converted from inches to the linear scale (Table 1). The model would be assessed by the ADGA Linear Appraisal Committee for comments and concurrence on the accuracy of the built in linear score scales.

Table 1: Measured Mammary Traits and Corresponding Linear Score:

Linear Score	Udder Depth Std (in)	Udder Depth Mini (in)	Medial Ligament Std (in)	Medial Ligament Mini (in)	Teat Diameter Std (in)	Teat Diameter Mini (in)	Teat Length Std (in)	Teat Length Mini (in)
5	≤ -2	≤ -1	≤ -1	≤ -0.5	≤ ½	≤ ¼	≤ 1.0	≤ 0.5
10	-1	-0.5	-0.5	-0.25	1.0	0.5	1.5	0.75
15	0	0	0	0	1.5	0.75	2.0	1.0
20	1	0.5	0.5	0.25	2.0	1.0	2.5	1.25
25	2	1	1	0.5	2.5	1.25	3.0	1.5
30	3	1.5	1.5	0.75	3.0	1.5	3.5	1.75
35	4	2	2	1	3.5	1.75	4.0	2.0
40	5	2.5	2.5	1.25	4.0	2.0	4.5	2.25
45	6	3	3	1.5	4.5	2.25	5.0	2.5
50	≥ 7	≥ 3.5	≥ 3.5	≥ 1.75	≥ 5.0	≥ 2.5	≥ 5.5	≥ 2.75

To address Aim 2 and assess the efficacy of the model on scoring accuracy we will conduct the following experiments:

Experiment 1: Appraisers/former appraisers (minimum of 3) would be asked to score rear udder traits on 20 animals. Each animal would be breed identified and have a photo taken of the rear udder with a ruler for scale to submitted to the research team with the scores assigned by the appraiser. The team would then assess the agreement between the photo provided and the morphology of the model created by the assigned scores as a proxy for appraiser accuracy.

Experiment 2: Each appraisers/former appraisers (minimum of 3) would be given an identical set of 20 udders images, with a breed and size scale included, randomized and asked to assign linear scores. They would submit those scores to the team. They would then be given the tool and asked to enter their scores into the tool and then adjust their scores based on morphologic image created by the tool. The adjusted scores would then be submitted to the team for analysis of inter-appraiser agreement before and after using the tool.

Experiment 3: Appraisers/former appraisers (minimum of 3) would be asked to score live animals (20 head per appraiser) in real time using the tool as an aid. They would submit a photo

of the udder with ruler for scale, their original assigned scores, as well as any adjusted scores after they entered their scores into the tool.
For all experiments data analysts would be blinded to which scores were given before or after use of the tool.

This collaboration will serve as a proof-of-concept for morphometric model that could be integrated into an app-based program for field use in which appraisers would receive an immediate visual rendering for side-by-side comparison with the live animal, creating a structured feedback loop. The pilot evaluation will compare scoring outcomes with and without the tool, assessing changes in score distribution and inter-appraiser consistency. Completion of these aims will generate preliminary data supporting submission to competitive extramural funding programs including USDA-NIFA AFRI in Animal Breeding and Genetics and ADGA Research Foundation focused on animal breeding, precision phenotyping, and data-driven decision systems. For the Smith Lab, this collaboration directly supports our mission to improve dairy goat sustainability through stronger, more biologically meaningful genetic selection tools. By generating pilot data that improves the accuracy and consistency of conformation scoring, this project strengthens the reliability of phenotypic data that underpin breeding and management decisions.
Importantly, this aligns with my broader research focus on dairy goat production and health and the use of existing data sources (e.g., USDA, NAHMS, and ADGA) to evaluate factors affecting milk production, milk quality, and functional longevity, ultimately enhancing the value and interpretability of these datasets.

Data: All of the data required for model development is outlined in the proposal above and is included in the 2025 ADGA Linear SOP and the 2019 ADGA Linear Appraisal Blue Book. For the data analysis of field testing, we have included an example of the worksheet we use to assess inter-appraiser variability during in person training on live animals (Table 2). We would use similar format for data collection from our appraisers during our scoring experiments. Additional I have included two photos, one standard and one miniature goat with a scale marker (Figure 4) that we would expect to receive so we could apply the scoring rubric to the photo and compare those scores to the scores given by the appraiser, entering them both into the model and assessing the outcome.

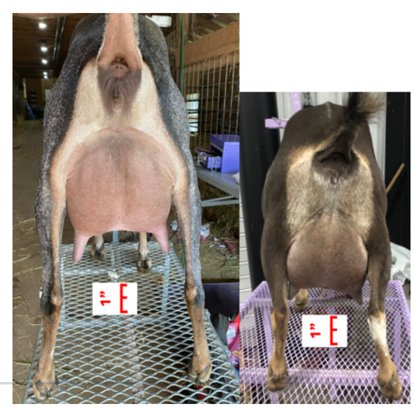

Table 2: Example scoring sheet

APPRAISER ID	ANIMAL ID	Breed	Age (Y)	Lactation #	Rear Udder Height	Rear Udder Arch	Medial Suspensory Ligament	Udder Depth	Teat Placement Rear View	Teat Diameter	Teat Length
27	10085	LM	3	3	28	38	20	26	23	24	24
22	10085	LM	3	3	27	38	26	27	23	29	23
23	10085	LM	3	3	26	38	22	30	22	30	22
25	10085	LM	3	3	31	40	20	28	24	24	25
18	10085	LM	3	3	25	27	22	30	22	23	26
24	10085	LM	3	3	35	40	20	27	26	27	23
			MEDIAN		27.5	38	21	27.5	23	25.5	23.5
			MINIMUM		25	27	20	26	22	23	22
			MAXIMUM		35	40	26	30	26	30	26
			RANGE		10	13	6	4	4	7	4
			MEAN		28.7	36.8	21.7	28.0	23.3	26.2	23.8
			ST DEV		3.7	4.9	2.3	1.7	1.5	2.9	1.5
			CO VARIATION		13.0	13.3	10.8	6.0	6.2	11.2	6.2

Team overview:

Principle Investigator(Bio sketch Below):

Dr Fauna Smith DVM PhD DACVIM, Assistant Professor of Livestock Herd Health and Reproduction, Department of Population Health and Reproduction, Weill School of Veterinary Medicine, University of California, Davis. flsmith@ucdavis.edu

Members:

Nora Manring, Current Animal Biology Undergraduate, Admitted to Animal Biology Graduate Group Starting Fall 2026, University of California, Davis ezmanring@ucdavis.edu

Ben Rupchis, Unit Manager, UC Davis Goat Facility, Department of Animal Science and Former American Dairy Goat Association Appraiser, barupchis@ucdavis.edu

Trinity Malmanis, Chair, Linear Appraisal Committee, American Dairy Goat Association and Former American Dairy Goat Association Appraiser, trinity@goatsan.com

Availability: This project is ready to go for model development ASAP; however, the testing phase would be better suited to summer. Goats are seasonal animals and in spring many of them have recently given birth. Our peak scoring season is summer, so from a model testing perspective, data gathering would be preferable in the summer. If the 8-12 weeks must be continuous then summer would be better, but if we could do model development in spring and testing in summer that would also work.

BIOGRAPHICAL SKETCH

NAME	POSITION TITLE
Fauna L Smith DVM PhD DACVIM	Assistant Professor Livestock Reproduction and Herd Health
flsmith@ucdavis.edu	Department of Population Health and Reproduction
	School of Veterinary Medicine
	University of California, Davis

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE	MM/YY	FIELD OF STUDY
University of California, Davis	PhD	05/22	Integrative Pathobiology
University of California, Davis	DVM	06/05	Veterinary Medicine
Carleton College	BA	06/01	Biology

Positions and Employment

1995-1997	Hammerich's Ranch Grade B Goat Dairy, Fulton, CA, Milker
1997-2000	Redwood Hill Grade A Goat Dairy and Creamery, Sebastopol, CA, Animal Health Manager
2001-2001	Creekside Cow Dairy, Escalon, CA, Interim Herd Manager
2005-2010	Matamata Veterinary Services Limited, Matamata, New Zealand, Mixed Livestock/Equine Veterinarian
2010-2014	Equine Veterinary Stud Service Limited, Matamata, New Zealand, Owner
2014-2017	Large Animal Internal Medicine, University of California, Davis William R Pritchard Veterinary Medic, Davis, CA, Resident

Other Experience and Professional Memberships

1999-2006	Licensed Judge - American Dairy Goat Association

2018	Member - Genetic Advancement Committee, American Dairy Goat Association
2018-2026	Diplomate of American College of Veterinary Internal Medicine (DACVIM)
2018-2026	Member - Reproductive Technologies Committee, American Dairy Goat Association
2019-2026	Chair - Genetic Advancement Committee and Subject Matter Expert, American Dairy Goat Association
2022-2026	Member - UC Davis Goat Day Organizing Committee
2024-2026	Chair - American Association of Small Ruminant Practitioners/American Dairy Goat Association Joint CE Meeting Committee
2024-2026	Member - Linear Appraisal Committee, American Dairy Goat Association
2024-2026	Member - Production Testing Committee, American Dairy Goat Association
2024-2026	Member - Small Ruminant Clinical Practice Representative, American Veterinary Medical Association, Committee on Environmental Issues

Selected peer-reviewed publications:

1. Smith FL, Heller MC, Crossley BM, Clothier KA, Anderson ML, Barnum SS, Pusterla N, Rowe JD. Diarrhea outbreak associated with coronavirus infection in adult dairy goats. J Vet Intern Med, 36(2): 805-811, 2022.
2. Smith FL, Savage HP, Luo Z, Tipton CM, Lee FE, Apostol AC, Beaudin AE, Lopez DA, Jensen I, Keller S, Baumgarth N. B-1 plasma cells require non-cognate CD4 T cell help to generate a unique repertoire of natural IgM. J Exp Med, 220(4): e20220195, 2023.
3. Mercer MA, Zhang Z, Clapham MO, Wetzlich SE, Smith FL, Rupchis BA, Lin Z, Tell LA. Plasma pharmacokinetics, milk residue depletion profile, and milk withdrawal interval estimation following multiple-dose oral administration of meloxicam to lactating dairy goats. Front Vet Sci, 12: 1620476, 2025.
4. Garzon A, Kell T, McNabb B, Keel K, Smith F. Bilateral uterine horn segmental aplasia in a doe. Clinical Theriogenology, 2025.
5. Smith FL, Fan F, Woods-Cuneo S, Depenbrock S. High-Mortality Outbreak of *Staphylococcus aureus* Mastitis Associated with Poor Milking Practices in a Goat Dairy. Veterinary Science,*13*(2), 203, 2026.

Ongoing Research Support

CALV-AH-449 10/1/2024 - 6/30/2026
Center for Food Animal Health
Molecular detection of SRLV in nasopharyngeal swabs of sheep
Role: Principal Investigator

CA-V-PHR-4159-H 10/1/2024 - 06/30/2026
Center for Food Animal Health
Characterization of hematopoietic chimerism and fertility in goats
Role: Principal Investigator

Sharif Aly (PI) 10/1/2024 - 6/30/2026
Center for Food Animal Health
Effect of supplementation of Zinc and Biochar on growth, health, immunity and mortality in pre-weaned dairy

Role: Co-Principal Investigator

Completed Research Support

MC Heller (PI) 06/30/2021 - 06/30/2022
Center for Food Animal Health
Safety and Immunogenicity of Bovine Coronavirus Vaccines in Goats
Role: Co-Principal Investigator

CALV-AH-430 10/1/2022 - 6/30/2024
Center for Food Animal Health
Field trial to evaluate the effect of a Mannheimia hemolytica/Pasteurella multocida vaccine on respiratory disease incidence and growth in lambs
Role: Principal Investigator

CALV-AH-442 10/1/2023 - 6/30/2025
Center for Food Animal Health
Validation of pooled fecal samples for MAP detection in small ruminants.
Role: Principal Investigator

CA-V-PHR4143-H 10/1/2023 - 6/30/2025
Center for Food Animal Health
Characterization of circulating strains of caprine coronavirus
Role: Principal Investigator